Subset Sums in a Fourfold Sumset

Joint covering losses and the audited bound $N \le 95$

Sum-of-Subset research project

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Abstract

We exclude cardinality 96 and extend the independently audited chain to $71 \le N \le 95$. A sharp four-extreme estimate reduces two coupled budgets by two. The covering is then organized by actual union deficits: leading twofold translates and retained onefold translates share one loss budget. This handles three empty units and multiple large-loss centers without a global "at most one large loss" assumption. Positive low holes are transported to 14; 1,241 retained oriented high-hole states are compressed into 26 endpoint groups. All endpoint certificates are newly generated and checked by exact integer arithmetic. The stronger archived bound 84 is preserved, but its entire proof chain has not been independently replayed; this result advances the audited chain and does not surpass 84.

1 Statement and inherited foundation

For finite $A \subset \mathbb{R}$, write $\mathrm{FS}(A) = \{\sum_{x\in X} x : X \subseteq A\}$, including the empty sum, and $4A = A+A+A+A$, allowing repeated summands. Let N be the maximum cardinality under $\mathrm{FS}(A) \subseteq 4A$.

Theorem 1.1. *Every feasible set has at most 95 elements. Hence* $71 \le N \le 95$.

The lower bound is the unchanged construction $[-31, 31] \cap \mathbb{Z} \cup \{\pm 93, \pm 155, \pm 217, \pm 713\}$. The upper-bound foundation is the independently audited chain $115 \to 103 \to 100 \to 99 \to 98 \to 97 \to 96$; see [1, 2] and their dependencies. We reuse its integralization, fixed covering, peeling, inverse, effective-hole, span, mass, and pairing lemmas. We reprove here the additional geometric inequalities drawn from the archived research report [3], and generate fresh certificates. No archived endpoint certificate is read.

A size-96 counterexample contains zero: otherwise adjoining zero gives a forbidden size-97 example. This uses the preceding global bound; feasibility is not assumed hereditary under arbitrary deletion. Integralize, remove four outer elements on each side to obtain D, and peel two largest and two smallest elements twice, obtaining $D \supset D_1 \supset E = P \cup \{0\} \cup (-Q)$. Put

$$d = 88, \quad e = 80, \quad m = a + b = 79, \quad a = |P|, \quad b = |Q|,$$
$$I = (a - 39)(a - 40), \quad T_r = r(r + 1)/2,$$
$$\Lambda = 1 + T_a + T_b = 1601 + I, \qquad \Omega = 22d + 97 - \Lambda = 432 - I.$$

2 A complete lattice gate at cardinality 96

For a positive r-element set X, define $\mu(X) = |\mathrm{FS}(X)| - 1 - T_r$ and $\rho(X) = \mu(X) + 2 - 2\delta_0(X)$, where $\delta_0(X)$ counts the x whose double is absent from the distinct-element two-sum set of $X \cup \{0\}$. Write $\mu = \mu(P) + \mu(Q)$ and $\rho = \rho(P) + \rho(Q)$. With the inherited nonnegative quantities,

$$\rho + 2\vartheta + c + \mathcal{T} = \Omega, \quad c = 2|2D| + 18d + 85 - |\mathrm{FS}(D)|, \quad \mathcal{T} = \kappa + \ell_1 + \ell_2 + g. \tag{1}$$

Here $\kappa = |\mathrm{FS}(E)| - \Lambda - \mu$; the ℓ_i are peeling slacks and $g = 4d - 10 - (|R(D)| - |R(D_1)|)$. The deficient-sign case is impossible since $1 + T_{80} > 24 \cdot 96 + 77$. The paired bootstrap gives $q^2 \le 2353$,

q odd, hence $q \le 47$ and $a, b \ge 16$. Nonnegative Ω leaves exactly $a = 19, \dots, 60$. For these 42 oriented pairs,

$$T_{\min(a,b)} - 2 - \Omega \ge 99 > 0.$$

The simultaneous inverse theorem therefore gives a common core lattice, with side-support holes satisfying

$$h_P, h_Q \le \kappa, \quad z = h_P + h_Q - \kappa, \quad 0 \le z \le \min(h_P, h_Q) \le \kappa, \quad h_X < |\operatorname{FS}(X)| - 1.$$

It remains to put the peeled elements on this lattice. The universal span bound $\max X \le r + \rho(X) + h_X + 2$, used before asserting lattice membership of the peeled layers, gives, for $s = |\operatorname{FS}(E)|$ and $C = 2m + 4e + 14 - \Lambda = -1109 - I$,

$$\begin{aligned} |\operatorname{FS}(D_1)| - 2s &\le C + \Omega + \rho - \mu + 2z, \\ |\operatorname{FS}(D_1)| - 2s &\le C + 3\Omega - \rho - \mu. \end{aligned}$$

The maximum is always defective, so $\rho \le \mu$. For $z \le 14$ the first bound is at most $-649 - 2I < 0$. For $z \ge 15$ both supports are non-progressions, so $\rho + \mu \ge (79 - 28) + (79 - 6) = 124$. The second bound is then $63 - 4I$. Any remaining pair has $I = k(k+1)$ with $k \le 3$, hence $a, b \ge 36$. The effective-hole row $(r, C_0, \beta) = (36, 109, 7)$ gives $\mu(P) \ge a + 110$, $\mu(Q) \ge b + 110$; its strict tests are checked exactly and remain valid for larger r. Thus $\rho + \mu \ge 350$, whereas $C + 3\Omega = 187 - 4I$. Across all pairs the resulting high-hole comparison is at most -17. Finally $\Lambda - (4d - 10 + \Omega) = 827 + 2I > 0$. The inherited overlap argument puts D on the core lattice and propagates its support holes. Only now do we use integer interval counts.

3 A sharp four-extreme defect inequality

Theorem 3.1. *Let $F = \{x_1 < \cdots < x_m\} \subset \mathbb{Z}$, $m \ge 7$, and let*

$$a_1 < a_2 < x_1 < \cdots < x_m < b_2 < b_1, \qquad D = F \cup \{a_1, a_2, b_2, b_1\}.$$

Let η_F, η_D count the holes in the endpoint hulls of $R(F), R(D)$, and put $g_H = 4m + 6 - (|R(D)| - |R(F)|)$. Then

$$\boxed{\eta_D + \eta_F + g_H \ge 2m - 2.} \tag{2}$$

The constant is sharp for every $m \ge 7$.

Proof. Write $L = x_1 + x_2$ and $U = x_{m-1} + x_m$. Let W_+ be $((b_1 + F) \cup (b_2 + F)) \cap (U, \infty)$, and let W_- be the corresponding union of the two lower translates below L. Put

$$I_+ = (U, b_1 + x_m] \cap \mathbb{Z}, \quad I_- = [a_1 + x_1, L) \cap \mathbb{Z}, \quad H_\pm = |I_\pm| - |W_\pm|.$$

Let q count the distinct mixed sums $a_i + b_j$ that lie outside $[L, U]$ and are not already in $W_- \cup W_+$. These sums lie in $I_- \cup I_+$, so $0 \le q \le 4$ and $q \le H_- + H_+$.

Translate x_1 to zero and write

$$\begin{aligned} x_m = M, \quad x_2 = s_0, \quad x_{m-1} = M - t_0, \\ a_2 = -w_0, \quad a_1 = -w_0 - v_-, \quad b_2 = M + u, \quad b_1 = M + u + v_+. \end{aligned}$$

All six gap parameters are positive integers. At most $u + 1$ members of $b_2 + F$ lie above U: at most u eligible points are at or below x_{m-1}, followed by x_m. Thus the integers

$$S_+ = u + 1 + m - |W_+|, \qquad S_- = w_0 + 1 + m - |W_-|$$

are nonnegative. Set $H = H_+ + H_-$ and $S = S_+ + S_-$. We claim

$$H + S \ge 2q - 2. \tag{3}$$

For $q \le 2$ this follows from $H \ge q$. The same bound proves the case $q = 3, S \ge 1$ and the case $q = 4, S \ge 2$.

Suppose first that $S = 0$. Since $|W_\pm| \le 2m$, we have $u, w_0 \le m - 1$. The mixed value $B_0 = a_2 + b_2 = M + u - w_0$ satisfies

$$M - m + 2 \le B_0 \le M + m - 2.$$

Now $s_0 \le M - m + 2$ and $x_{m-1} \ge m - 2$, so $B_0 \in [s_0, 2M - t_0] = [L, U]$. In particular $q = 4, S = 0$ is impossible.

If $q = 3, S = 0$, the other mixed values

$$A_0 = B_0 - v_-, \qquad C_0 = B_0 + v_+ - v_-, \qquad D_0 = B_0 + v_+$$

are all outside $[L, U]$. Thus $A_0 < L$ and $D_0 > U$. Reflection allows us to assume $C_0 < L$. The definitions give

$$H_+ = v_+ + t_0 - m - 1, \qquad H_- = v_- + s_0 - m - 1.$$

Using integrality in $D_0 > U$ and $C_0 < L$ yields, respectively,

$$u - w_0 \ge M - H_+ - m, \qquad u - w_0 \le t_0 - M + H_- - H_+ - 1.$$

Consequently

$$H_- \ge 2M - m - t_0 + 1 = M + x_{m-1} - m + 1 \ge M - 1 \ge m - 2 \ge 5.$$

This proves (3) in that case.

Finally, suppose $q = 4, S = 1$. Reflect if necessary so that $S_+ = 1, S_- = 0$. The cardinality bounds now give $u \le m$ and $w_0 \le m - 1$, so $B_0 \ge M - m + 2 \ge s_0$. Since B_0 lies outside the old hull, $B_0 > U$, whence

$$u - w_0 > x_{m-1} \ge m - 2, \qquad u - w_0 \le m - 1.$$

Equality forces $u = m$, $w_0 = 1$, and $x_{m-1} = m - 2$. The first $m - 1$ members of F are therefore $0, 1, \ldots, m - 2$, so $s_0 = 1$, and $B_0 = M + m - 1 = U + 1$. Because $v_- \ge 1$, the value $A_0 = B_0 - v_-$ is at most U; it must consequently be less than 1. Thus $v_- \ge M + m - 1$ and

$$H_- = v_- + s_0 - m - 1 = v_- - m \ge M - 1 \ge m - 2 \ge 5.$$

Together with $S = 1$, this proves (3) in the final case.

Put $h_- = L - (a_1 + a_2)$, $h_+ = (b_1 + b_2) - U$, and $h = h_- + h_+$. The definitions give the exact identity

$$2(|W_-| + |W_+|) = h + 2m + 2 - H - S.$$

Every new restricted sum outside $[L, U]$ lies in $W_- \cup W_+$, is one of the q new mixed values, or is one of $a_1 + a_2, b_1 + b_2$. The latter values are distinct and outside $I_- \cup I_+$. Hence the number N_{out} of these sums satisfies

$$2N_{\text{out}} = h + 2m + 6 - H - S + 2q \le h + 2m + 8.$$

Write $N_0 = |R(D)| - |R(F)|$. At most η_F new sums are inside the old hull, so $N_0 \le N_{\text{out}} + \eta_F$. Also $\eta_D = h + \eta_F - N_0$. Therefore

$$\begin{aligned} \eta_D + \eta_F + g_H &= h + 2\eta_F + 4m + 6 - 2N_0 \\ &\ge h + 4m + 6 - 2N_{\text{out}} \ge 2m - 2. \end{aligned}$$

For sharpness, take $F = [0, m - 1]_{\mathbb{Z}}$ and $\{a_1, a_2, b_2, b_1\} = \{-m - 2, -1, m, 2m + 1\}$. Here $R(F) = [1, 2m - 3]_{\mathbb{Z}}$ and $R(D) = [-m - 3, 3m + 1]_{\mathbb{Z}}$. For the latter assertion the four translates of F fill the ranges $[-m - 2, -3]$, $[-1, m - 2]$, $[m, 2m - 1]$, $[2m + 1, 3m]$; $R(F)$ fills the central gap, and the sums of outer elements supply $-m - 3, -2, 2m, 3m + 1$. Thus both hole counts vanish and $g_H = 2m - 2$. □

4 Coupled budgets and endpoint transport

Apply the sharp theorem to $F = D_1$, so $G = 2(d-4) - 2 = 166$. The inherited hole propagation $w \le 4z$ yields

$$\mathcal{T} \ge \kappa + G - 3z, \qquad \mathcal{T} \ge \max(56, z, 166 - 2z). \tag{4}$$

If j of the eighteen covering units are empty, put $\delta = c - 88j$. Combining the span lemma with (1)–(4) gives

$$R_p + R_q + \delta \le 349 - I - 88j + 4z, \qquad R_p = \max P, \quad R_q = \max Q. \tag{5}$$

For $z = 0$, $c \le 266 - I$; for $1 \le z \le 14$, both supports are non-progressions and $\rho \ge 51$, so $c \le 215 - I + 2z \le 243 - I$. Consequently the complete low-hole root ranges are

Regime	Z_0	Z_1	Z_2	Z_3	P_0	P_1	P_2
a	24–55	27–52	30–49	38–41	24–55	28–51	32–47
Integer roots	128	1820	11560	11968	128	1680	9248

These ranges deliberately retain every parameter allowed by the scalar cap. In particular, Z_1 allows $\delta \le 178 \ge 2 \cdot 84$ and Z_2 allows $\delta \le 90 \ge 84$. Neither "one large loss" nor universal Z_2 core quantiles may be assumed. In Z_3, $\delta \le 2$, so ordinary quantiles are valid. In P_2, $\delta \le 67$, so core quantiles are valid.

For $z \ge 15$, the checker reconstructs the inherited effective-hole catalogue. For each side it chooses the strongest available certified $u = r + C_0 + 1$ with $2\beta < z$, and checks the strict effective-hole tests, including the endpoint bridge. The flexible inverse gives

$$\rho_*(r, u) = \max\{r - 14, u + 2 - 2r, L_2(r, u), \dots, L_{10}(r, u)\},$$

where, with $v = \max(0, u - 4\beta + 2) = (\beta + 1)q + s$,

$$L_\beta(r, u) = \min\{(\beta - 1)(r - 14), (\beta - 1)q + \min(s, \beta - 1)\}.$$

Thus $c \le \Omega - \rho_*(a, u_p) - \rho_*(b, u_q) - \max(56, z, 166 - 2z)$. There are 2,840 surviving oriented scalar states and $c \le 144 < 176$, so $j \le 1$. For $j = 0$, first orient the actual first-pair contributor positively. In that fixed orientation the active bound is one of

$$F \ge \tfrac{1}{2}T_a + T_b + 6a + 8b + 85 + \tfrac{1}{2}(u_p + z) + u_q + z,$$
$$F \ge \tfrac{2}{3}T_a + T_b + 6a + (22b + 236)/3 + \tfrac{2}{3}(u_p + z) + u_q + z,$$

while $F \le 1710 + 4z$. A positive minimum excludes that oriented state; choosing the stronger reflected orientation is invalid. This leaves 477 states for $j = 0$ and 764 for $j = 1$. For H_1, $\delta \le 56 < 84$ permits core quantiles.

Lemma 4.1 (Mass transport)**.** *For fixed side sizes and common lower bounds u_p, u_q, let Z be the largest hole count in a group and C_* the largest covering cap. Replace S_p, S_q by $S_p + Z - z, S_q + Z - z$, leave all other coordinates unchanged, and replace z by Z. Every necessary row used below continues to hold with cap C_*.*

Proof. Writing $t = Z - z \ge 0$, the doubled mass row has change $-2t + 2t = 0$, the exposed mass row has change $-t + t = 0$, the cardinality row has change $2t - 4t = -2t$, and the span row has change $-4t$. Upper quantiles and leading-slot upper rows only weaken. All overlap rows are unchanged. The low cap $\delta - 2z$ decreases by $2t$; the high cap is replaced by its maximum. Nonnegativity is preserved. The shifted masses are auxiliary coordinates, not a claim about a new actual set. All shifts are integral. □

For low positive holes take $Z = 14$. For high holes group by (a, j, u_p, u_q), regardless of which affine part of the scalar cap is active. This yields 12 H_0 and 14 H_1 groups, hence $12 \cdot 4 + 14 \cdot 70 = 1028$ integer models instead of $477 \cdot 4 + 764 \cdot 70 = 55388$. The checker compares every source row with its transported row on all 1,241 retained states. This is in addition to the general coefficient argument above.

5 One loss budget for overlapping covering slots

Let $S_p = \sum(D \cap \mathbb{R}_{>0})$ and $S_q = -\sum(D \cap \mathbb{R}_{<0})$, initially before transport. Write the original outer magnitudes as $p_1 > p_2 > p_3 > p_4 > M_p$ and their negative-side analogues q_i. Put $c_p = p_1 - p_2 - p_3 - p_4$, $\alpha_p = p_2 - p_3 - p_4$, $\beta_p = -p_1 + 2p_2 - p_3 - p_4$, and reflect for q. The twenty center labels are

$$v_{p,i} = c_p + p_{i+1}\ (0 \le i \le 3), \quad v_{p,i+4} = c_p - q_{i+1}\ (0 \le i \le 3), \quad v_{p,8} = \alpha_p, \quad v_{p,9} = \beta_p,$$

and their reflected counterparts. Pair 1 is $p4, q4$. In branch A, $\beta_p + \beta_q \le 0$, pair 2 is $p9, q9$; in branch B it is $p5, q5$. There are sixteen singleton units and two paired units, each of capacity d. Enumerate every j-subset of missing units. If pair 1 is present, orient its actual contributor positively. If pair 2 is present and not potentially split, select either actual contributor. In branch A with $p5$ or $q5$ missing, pair 2 can split: do not count an individual loss for it, but select an actual member and retain $\beta_s + M_s \ge 1$. The numbers of modes for $j = 0, 1, 2, 3$ are $4, 70, 578, 2992$; both member choices are retained.

For each retained individual center let l_v, u_v be its point counts below and above the target integer hull, and $\lambda_v = l_v + u_v$. Let $K = |2D|$ and let L_p, L_q be the losses of the leading slots $c_p + 2D, c_q + 2D$ from their actual positive or reflected targets. The fixed covering gives

$$L_p + L_q + \sum_v \lambda_v \le \delta. \tag{6}$$

We strengthen this by replacing groups of slots by their actual union deficits. This construction is valid even when a leading slot is empty.

Proposition 5.1 (Ordinary and leading groups)**.** *For k retained same-sign individual slots with extreme centers $v_{\min}, v_{\max}$, their union deficit is at least*

$$H_G + \max_v \lambda_v, \qquad H_G = ke - (R_p + R_q + v_{\max} - v_{\min} + 1),$$

and at least $\sum_v \lambda_v$. For a leading slot and k retained own-center slots $c_p + p_i + D$, the union deficit is at least

$$H_* + \max(L_p, \lambda_1, \ldots, \lambda_k), \qquad H_* = (k+2)e - 2 - p_{\max} - R_p - 2R_q,$$

and at least $L_p + \sum \lambda_i$. Reflect on the other side.

Proof. The core pieces of the first group lie in an integer interval of length $R_p + R_q + v_{\max} - v_{\min} + 1$; the other points number at most $k(d-e)$. Subtract this raw union bound from kd. Points outside the target include each member's loss set, adding at least its largest loss. The sum bound follows by bounding the union by the sum of the individual target contributions. For the second group, $c_p + 2E$ and the $c_p + p_i + E$ lie in $[c_p - 2R_q, c_p + p_{\max} + R_p]_{\mathbb{Z}}$. Since $|2E| \ge 2e - 1$, the other points number at most $K - (2e-1) + k(d-e)$. Subtraction from $K + kd$ gives H_*. The same outside-target and sum arguments finish the proof. □

The leading centers are integral with at most one empty unit by the inherited contributor argument. With two or three empty units, $\delta \le 90 < 175 \le K$, so both leading slots contribute and their centers are integral. Thus the integer hull count in this proposition is justified in every regime.

The model partitions the retained centers, on each side, into own labels $0, 1, 2, 3$, cross labels $4, 5, 6, 7$, and ungrouped labels $8, 9$. Each own block also includes its leading slot; at most three missing units ensure that at least one own label remains. Introduce one nonnegative variable O for each own block and for each cross block of size at least two. Choose O to be that block's actual union deficit. Impose its sum-of-losses bound and its ordinary hull bound for each member. For every nonempty subset of an own block impose the leading bound for that subset, plus the losses of

the unselected members. All these rows are valid: a union deficit for disjoint groups of slot labels is at least the sum of their separate deficits. Finally,

$$\sum_{\text{blocks}} O + \sum_{\text{ungrouped } v} \lambda_v \le \delta.$$

No overlap gain is counted twice. The variables O represent actual whole-block deficits; alternative subset bounds are inequalities for the same variable, not separate credits added to the budget.

6 Remaining necessary rows and finite disjunctions

Write $x_0 = R_p < x_1 < x_2 < x_3 < x_4 = M_p$ and similarly y_i. All coordinates and losses are nonnegative. Besides the cap and span rows, the model contains the inherited mass rows, on each side:

$$\begin{aligned} 2M_p + 6R_p - 2S_p + 2z &\le -2T_a - 12 - 2u_p, \\ M_p + 4R_p - S_p &\le -T_{a-1} - 6, \\ x_1 + x_2 + x_3 + M_p - S_p + z &\le -T_a - u_p, \\ R_p + x_1 + x_2 + x_3 + M_p - S_p &\le -T_{a-1}. \end{aligned}$$

Here $u_p = 0$ for Z, $a-3$ for P, and the certified effective-hole bound for H; transport changes z to its endpoint and the mass constants accordingly. Also $R_p \ge a$, $M_p \ge R_p + 4$, all inner gaps are at least one, and

$$S_p + S_q + \delta - 2K - 4z \le 1668 - 88j.$$

Global positive-center quantiles are $v + l_v + M_p \ge 88$ and $v - u_v - M_q \le S_p - 87$. Core quantiles replace M_p, M_q by R_p, R_q and 88,87 by 84,83. Ordinary quantiles are $v+l_v \ge b+5$, $v-u_v \le S_p-a-4$. Use global quantiles generally, core quantiles for P_2, H_1, and ordinary quantiles for Z_3. Individual loss caps are 87 globally and 83 in core models. Reflection supplies the other side.

If $j \ge 2$, or the regime is P_1 or H, then $\delta < 167$. Counting the leading-slot contributors yields

$$\begin{aligned} &c_p + 2M_p + L_p \ge K, && S_p + 2M_q - c_p + L_p \ge K - 1, && S_p + L_p \ge K, \\ &c_p + 2x_i + L_p \ge 167 + 2i, && S_p + 2y_i - c_p + L_p \ge 166 + 2i. \end{aligned}$$

Indeed, a set consisting of E, four negative outliers and i positive outliers has at least $2(e+4+i)-1 = 167 + 2i$ twofold sums; at most L_p fail to contribute. The bound $L_p \le \delta < 167$ ensures a nonempty counted interval. The three K rows follow by counting all $K - L_p$ contributors. The coarse model replaces K here by its lower bound 175.

With two empty units the leading centers are integral. If an original outer element were nonintegral, its own unpaired slot would be empty. There cannot be exactly one such element on a side, since c_p or c_q is integral. Exactly two would exhaust the missing units; the unpaired cross labels force outer elements 3 and 4 integral, and in branch A the remaining cross label forces element 2 integral, while in branch B the unpaired α does so. The leading-center identity then forces element 1 integral, a contradiction. Thus strict outer gaps and $\beta_p + \beta_q \ge 1$ in branch B are valid for $j \le 2$. For $j = 3$ only weak outer gaps and a weak branch-B inequality are used.

The coarse envelope is $K \le M_p + M_q + R_p + R_q + 21$; substitute it into the cardinality row to eliminate K. Only the remaining difficult groups retain inner coordinates and the five nested envelopes

$$K \le 2(x_t + y_t) + 1 + T_{2(4-t)} + \sum_{s \in \{p,q\}} \sum_{i=t+1}^{4} g_{s,t,i},$$

where $0 \le g_{p,t,i} \le x_i - x_{i-1}$ and $g_{p,t,i} \le x_t + y_t + 1$, and similarly for q. These are the inherited interval-union envelope rows. We also impose

$$K \le M_p + M_q + x_i + y_h + 1 + T_{4-i} + T_{4-h} \quad (0 \le i, h \le 4).$$

Every mixed-sign sum and every same-sign sum with a summand below its cutoff lies in $[-M_q - y_h, M_p + x_i]$. The only exceptions are pairs among the remaining outliers, at most the displayed triangular counts. This proves all 25 additional envelopes.

The finite trees use only the following exhaustive alternatives. All statements apply equally after mass transport, with the enlarged upper endpoint used to select an upper-endpoint alternative.

1. *Core or large loss:* either $l_v + u_v \geq 84$, or $l_v + u_v \leq 83$ and both core quantiles hold. Actual losses are integers; in the second case more than four points contribute, giving the core counting inequalities. No bound on the number of large centers is needed.

2. *Ordinary or outside:* either $v + l_v \geq b + 5$, or $l_v \geq b + 5$ and $v \leq -1$. At the upper end either $v - u_v \leq S_p - a - 4$, or $u_v \geq a + 5$ and $v \geq S_p + 1$. For nonnegative v, count the $b + 5$ nonpositive members; for negative v all are lost. At the upper endpoint use the $a + 4$ positive members and, outside the hull, zero. Integral centers justify the unit gaps. Enlarging the endpoint decreases the actual upper loss, so the original loss variables still dominate the necessary counts.

3. *Inner endpoint:* for $1 \leq i \leq 4$, either $l_v \geq i$ or $v \geq y_{5-i} + 1$; either $u_v \geq i$ or $v \leq S_p - x_{5-i}$. Otherwise all of the corresponding i extreme points are lost.

4. *Both or sole pair contributor:* in a split pair with selected actual member s, either the other member satisfies the necessary condition $\beta_t + M_t \geq 1$, or s is the sole contributor. In the latter case introduce its two hull losses l_*, u_*, impose their global quantiles, and add them to the sum-of-individual-losses budget. Only necessity is used in the first branch; positivity alone is never asserted sufficient for contribution.

All alternatives keep both children. Since transformed masses, centers and actual losses are integral, the displayed integer gaps remain valid. Tree variables themselves may be relaxed to real nonnegative values.

7 Exact certificate coverage and reproducibility

The verifier reconstructs the complete root set from the scalar gates and mode grammar. A root uses either the coarse model or the full nested model. It does not enumerate designated large-loss positions: where necessary, the proved core-or-large binary split supplies that information locally. In the final certificate bundle:

Quantity	Count
Structural root groups	5,324
Integer root systems	37,560
Groups using the coarse model	5,301
Groups using the nested model	23
Direct symbolic interval checks	5,350
Integer roots requiring a tree	141
Binary internal nodes / tree leaves	275 / 416
Maximum tree depth	5
Final integer leaf systems	37,835
Distinct integer certificate vectors	651

The 275 splits consist of 207 core-or-large, 43 ordinary-or-outside, 24 both-or-sole pair, and one inner-endpoint split. The count identity $416 = 141 + 275$ is verified. The minimum normalized strict certificate gap is $1/73$.

For each system $Ax \leq b$, $x \geq 0$, a certificate is an integer row weight vector $w \geq 0$ with $w^T A \geq 0$ coefficientwise and $w^T b < 0$. Then $0 \leq w^T Ax \leq w^T b < 0$, a contradiction. The normalized gap

is $-w^T b/\sum w_i$. In a parameter interval the matrix is fixed and the right-hand sides are directly constructed as quadratic polynomials in the integer side size a. Their weighted maximum is checked at both endpoints and the relevant integers around the vertex, using exact rational arithmetic. No polynomial is inferred from sampling. There are 13,020 such extremum-point checks.

Every integer parameter is covered exactly once by a direct interval or a complete two-child tree. Unknown rows, nonpositive contradictions, negative weights, missing parameters, missing roots, unknown splits, repeated splits on a path, and missing children are rejected. The 15 negative controls also reject omission of a three-empty mode, omission of a split pair's actual contributor, the invalid universal Z_2 core shortcut, the invalid one-large-loss shortcut, arbitrary reflection in the active estimate, and an unsupported claim about cardinality 95. In particular, this program certifies $N \leq 95$, not $N \leq 94$.

Files and commands. The verifier is `verification/upper95.py`.

Its certificate data are `data/upper95_certificates.json`.

Verification needs Python 3.10 or later and only its standard library:

```
python -I -S -B verification/upper95.py --output .work/upper95.json
python -I -S -B verification/upper95.py --exhaustive
```

The exhaustive mode regenerates every numerical integer model, compares every row with the symbolic formula, and checks every assigned certificate independently of interval compression. All tree leaves are always checked at their actual integer parameter. The optional generator `experiments/generate_upper95_certificates.py` uses SciPy to search for multipliers and useful disjunctions; floating-point solver status is not accepted as proof. Every saved multiplier is rechecked exactly.

The actual verification runs and hashes are stored under `results/2026-10-16/upper95/`. Standard and optimized Python outputs agree byte for byte; exhaustive and interval results agree apart from the mode label; regenerated certificates agree byte for byte; the preceding upper-96 result is unchanged. The Windows entry `verify_upper95_windows.bat` runs the seven finite continuations from 115 down to 95. Program success validates the finite certificates; the infinite mathematical reductions are supplied by the report and its cited foundation, rather than by a proof assistant.

What the method improves. Three-empty modes are handled explicitly with their tiny residual budget. All other modes use a common loss geometry, so increasing the allowed number of large-loss centers does not require a new global case grammar. Only 23 of the 5,324 structural groups need nested geometry; every other group closes with the coarse model. The high-hole transport independently removes about 98.1% of the unmerged high-hole model instances.

At the next cardinality 95, the balanced coarse parameters would be $d = 87$, $e = 79$, $\Omega = 450$, $G = 164$. The low positive-hole cap can reach $450 - 50 - 164 + 28 = 264 \geq 3 \cdot 87$, so positive three-empty modes must be addressed as well. No exclusion of that endpoint is claimed.

References

[1] Sum-of-Subset research project, *Transported hole bounds and a proof of $N \leq 96$*, `reports/upper96_from97_20261016.tex`, 16 October 2026.

[2] Sum-of-Subset research project, *Proof-method optimization*, `reports/proof_method_20261016.tex`, 16 October 2026.

[3] Sum-of-Subset research project, *Complete upper-86 research report*, `reports/upper86_complete_20261005.tex`, 5 October 2026. The sharp four-extreme, grouped-overlap, leading-slot and outer-integrality arguments are reconstructed here; its endpoint certificates are not used.